

# Sampling and Representation

Project: Transparent Book Recommender

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

### Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

### Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

## 2. Practise with fading help

Use W2: Read the sample result for Questions populations and samples. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

---

---

---

## 3. Individual reasoning check

Use the worked case for Questions populations and samples. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

---

---

---

---

## 4. Independent transfer case

## Independent case: Plan independently

Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

### My independent explanation, trace and limits

---

---

---

---

---

---

---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 responses may still fail to represent the school. Propose a better sampling plan using fictional data.

## 6. Solve the given case

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

---

---

---

---

---

## 7. Check a changed case

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My independent answer and reason

**Write or draw your reasoning and evidence.**

---

---

---

---

---

### Week 19 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

---

---

---

---

---

## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 responses may still fail to represent the school. Propose a better sampling plan using fictional data.

**My independently reasoned answer**

---

---

---

---

---

---

---

Record support used: none, hint or teacher explanation.

## 12. W1 Define the population

The question is whether all Grade 7 students prefer a longer lunch. Name the population and one unsuitable sample.

**My reasoning and evidence**

---

---

---

---

---

---

**13. W2 Read the sample result**

Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

**My reasoning and evidence**

---

---

---

---

---

---

**14. W3 Repair a claim**

A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

**My reasoning and evidence**

---

---

---

---

---

---

## 15. W4 Plan independently

Design an anonymous sample plan to ask all Grade 7 about three activity choices. State population, sampling frame, selection method, question, and how you will report nonresponse.

### My reasoning and evidence

---

---

---

---

---

---

---

---

## 16. W5 Separate invitation and response

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report nonresponse?

### My reasoning and evidence

---

---

---

---

---

---

## 17. Graphs and honest scales — W1 Read the scale

A bar is 6 squares long with 2 votes per square. How many votes is that?

**My reasoning and evidence**

---

---

---

---

---

---

## 18. Graphs and honest scales — W2 Compare fairly

Counts are 12 and 15. State the difference and ratio using the true values.

**My reasoning and evidence**

---

---

---

---

---

---

## 19. Graphs and honest scales — W3 Repair a misleading graph

Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

**My reasoning and evidence**

---

---

---

---

---

---



## 20. Graphs and honest scales — W4 Draw independently

Draw a bar chart for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero baseline and equal scale. State two supported comparisons.

### My reasoning and evidence

---

---

---

---

---

---

---

---

## 21. Graphs and honest scales — W5 Limit the claim

Art has the most votes among 18 responses. Does that establish Art is best for learning?

### My reasoning and evidence

---

---

---

---

---

---

---

## 22. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

---

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 23. Pictorial case practice: Sampling and Representation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

## Sampling and Representation

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Define target

The question concerns all Grade 7, not only library-club members.



#### 2. Select invitations

Teacher uses the complete approved list and documented random selection.



#### 3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



#### 4. Report scope

Invitation response rate  $12/15=80\%$ ; preference counts describe respondents.

### CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Can 80% response be reported as 80% of the whole grade responding?

### My picture-based explanation and changed-case prediction

---

---

---

---

---

---

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

## 24. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

---

---

---

---

---